Computer Science Mid Term Exam Notebook – Students Social Media Addiction

The digital era has transformed social media into a primary tool for communication and entertainment, but its excessive use among students raises concerns regarding mental health and academic performance.

This dataset provides a cross-sectional view of 705 students globally.

Your task is to analyze the data to extract some conclusions about behavioral patterns, addiction levels, and correlations between daily usage and lifestyle outcomes like sleep quality and relationship status.

Dataset Column Description

- Student_ID: Unique identifier for each participant.
- Age / Gender: Demographic profile of the student.
- Academic_Level: Educational tier (High School, Undergraduate, Graduate).
- Country: Geographical location of the student.
- Avg_Daily_Usage_Hours: Time spent daily on social platforms.
- Most_Used_Platform: The student's primary social network (e.g., TikTok, Instagram).
- Affects_Academic_Performance: Self-assessment of whether social media disrupts studies (Yes/No).
- Sleep_Hours_Per_Night: Average nightly sleep duration.
- Mental_Health_Score: Numeric scale representing mental well-being.
- Relationship_Status: Personal relationship status (Single, In Relationship, etc.).
- Conflicts_Over_Social_Media: Frequency of disputes caused by social media use.
- Addicted_Score: A metric quantifying the degree of addiction.

0. Set up the environment

We need to import the libraries and read the file.

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Read the file and show the first rows to check you have read data in a correct way.

```
df = pd.read_csv("Students Social Media Addiction.csv")
df.head()
```

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours	Most_Used_Platform	Affects_Academic_Performa
0	1	19	Female	Undergraduate	Bangladesh	5.2	Instagram	
1	2	22	Male	Graduate	India	2.1	Twitter	
2	3	20	Female	Undergraduate	USA	6.0	TikTok	
3	4	18	Male	High School	UK	3.0	YouTube	
4	5	21	Male	Graduate	Canada	4.5	Facebook	

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 705 entries, 0 to 704
Data columns (total 13 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Student_ID                    705 non-null    int64
 1   Age                           705 non-null    int64
 2   Gender                        705 non-null    object
 3   Academic_Level                705 non-null    object
 4   Country                       705 non-null    object
 5   Avg_Daily_Usage_Hours         705 non-null    float64
 6   Most_Used_Platform            705 non-null    object
 7   Affects_Academic_Performance  705 non-null    object
```

```
 8   Sleep_Hours_Per_Night         705 non-null    float64
 9   Mental_Health_Score           705 non-null    int64
 10  Relationship_Status           705 non-null    object
 11  Conflicts_Over_Social_Media   705 non-null    int64
 12  Addicted_Score                705 non-null    int64
dtypes: float64(2), int64(5), object(6)
memory usage: 71.7+ KB
```

How many rows and columns are there in the dataframe?

```
1
```

```
(705, 13)
```

1. Explore the dataset (3 points - 0,3 points for each question)

Q1 - Select rows from index 20 to 30

```
1
```

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours	Most_Used_Platform	Affects_Academic_Performa
20	21	18	Male	High School	Belgium	5.3	TikTok	
21	22	23	Female	Graduate	Switzerland	2.5	LinkedIn	
22	23	19	Male	Undergraduate	Austria	4.9	Instagram	
23	24	20	Female	Undergraduate	Portugal	5.7	TikTok	
24	25	22	Male	Graduate	Greece	3.2	Facebook	
25	26	19	Female	High School	Ireland	6.1	Instagram	
26	27	21	Male	Undergraduate	New Zealand	3.8	YouTube	
27	28	20	Female	Undergraduate	Singapore	4.4	TikTok	
28	29	24	Male	Graduate	Malaysia	2.2	LinkedIn	
29	30	19	Female	High School	Thailand	5.9	Instagram	
30	31	21	Male	Undergraduate	Vietnam	3.6	Facebook	

Q2. Select the rows where Gender is Female

```
1
2
3
```

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours	Most_Used_Platform	Affects_Academic_Perfor
0	1	19	Female	Undergraduate	Bangladesh	5.2	Instagram	
2	3	20	Female	Undergraduate	USA	6.0	TikTok	
5	6	19	Female	Undergraduate	Australia	7.2	Instagram	
7	8	20	Female	Undergraduate	Brazil	5.8	Snapchat	
9	10	21	Female	Graduate	South Korea	3.3	Instagram	
...	...	...	...	...	...	...	...	
696	697	21	Female	Undergraduate	Mexico	6.3	TikTok	
698	699	19	Female	Undergraduate	Canada	5.7	Instagram	
700	701	20	Female	Undergraduate	Italy	4.7	TikTok	
702	703	21	Female	Undergraduate	China	5.6	WeChat	
704	705	19	Female	Undergraduate	Poland	6.2	Facebook	

353 rows × 13 columns

Q3. How many students are younger than 21?

```
```

```
342
```

Q4. Create a new dataframe called df_older21 that incluees the columns `Age`, `Gender` and `Addicted_Score` for the students who are older than 21.

```
```

	Age	Gender	Addicted_Score
1	22	Male	3
6	23	Male	2
12	22	Male	4
16	24	Male	3
21	23	Female	4

Q5. How many students are male and how many are female?

```
```

```
Gender
Female    353
Male      352
Name: count, dtype: int64
```

Q6. Show the rows of the dataset in which the platform is Instagram or TikTok

```
```

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours	Most_Used_Platform	Affects_Academic_Perfor
0	1	19	Female	Undergraduate	Bangladesh	5.2	Instagram	
2	3	20	Female	Undergraduate	USA	6.0	TikTok	
5	6	19	Female	Undergraduate	Australia	7.2	Instagram	
8	9	18	Male	High School	Japan	4.0	TikTok	
9	10	21	Female	Graduate	South Korea	3.3	Instagram	
...	...	...	...	...	...	...	...	
694	695	20	Female	Undergraduate	Turkey	6.6	Instagram	
696	697	21	Female	Undergraduate	Mexico	6.3	TikTok	
698	699	19	Female	Undergraduate	Canada	5.7	Instagram	
700	701	20	Female	Undergraduate	Italy	4.7	TikTok	
701	702	23	Male	Graduate	Russia	6.8	Instagram	

403 rows × 13 columns

Q7. Show the students with Avg_Daily_Usage_Hours is between 4 and 7 hours

```
```

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours	Most_Used_Platform	Affects_Academic_Perfor
0	1	19	Female	Undergraduate	Bangladesh	5.2	Instagram	
2	3	20	Female	Undergraduate	USA	6.0	TikTok	
4	5	21	Male	Graduate	Canada	4.5	Facebook	
7	8	20	Female	Undergraduate	Brazil	5.8	Snapchat	
8	9	18	Male	High School	Japan	4.0	TikTok	
...	...	...	...	...	...	...	...	
700	701	20	Female	Undergraduate	Italy	4.7	TikTok	
701	702	23	Male	Graduate	Russia	6.8	Instagram	
702	703	21	Female	Undergraduate	China	5.6	WeChat	
703	704	24	Male	Graduate	Japan	4.3	Twitter	
704	705	19	Female	Undergraduate	Poland	6.2	Facebook	

516 rows × 13 columns

Q8. Show the students who use 'Instagram' and have an Addicted_Score equal or bigger than 8

```
1
```

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours	Most_Used_Platform	Affects_Academic_Perfor
0	1	19	Female	Undergraduate	Bangladesh	5.2	Instagram	
5	6	19	Female	Undergraduate	Australia	7.2	Instagram	
13	14	18	Female	High School	Mexico	6.5	Instagram	
17	18	19	Female	High School	Norway	5.0	Instagram	
25	26	19	Female	High School	Ireland	6.1	Instagram	
...	...	...	...	...	...	...	...	
667	668	24	Male	Graduate	Russia	6.9	Instagram	
677	678	23	Male	Graduate	Turkey	6.8	Instagram	
684	685	21	Female	Undergraduate	Russia	6.7	Instagram	
694	695	20	Female	Undergraduate	Turkey	6.6	Instagram	
701	702	23	Male	Graduate	Russia	6.8	Instagram	

63 rows × 13 columns

Q9. How many students see their performance affected (Affects_Academic_Performance) by social media?

```
1
2
3
```

```
453
```

Q10. Select the students who has a Mental_Health_Score lower than 5 and Sleep_Hours_Per_Night lower than 6.

```
1
2
3
```

2. Calculations and Transformations (3 points - 0,3 points for each question)

Q11. Calculate the average daily usage hours.

```
1
2
3
4
```

```
np.float64(4.918723404255319)
```

Q12. How many girls are there in the dataset who are 'In Relationship'?

```
1
2
3
```

```
Total de chicas en una relación: 138
```

Q13. Create a new column called 'High_usage' that will have a True value if 'Avg_Daily_Usage_Hours' is bigger than 5 hours.

```
1
```

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours	Most_Used_Platform	Affects_Academic_Performa
0	1	19	Female	Undergraduate	Bangladesh	5.2	Instagram	
1	2	22	Male	Graduate	India	2.1	Twitter	
2	3	20	Female	Undergraduate	USA	6.0	TikTok	
3	4	18	Male	High School	UK	3.0	YouTube	
4	5	21	Male	Graduate	Canada	4.5	Facebook	

Q14. Create a new column called 'High_Addiction' which will have a Yes value if the 'Addicted_Score' is equal and bigger than 7, otherwise it will have "No" value.

```
1
2
3
```

	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours	Most_Used_Platform	Affects_Academic_Performa
0	1	19	Female	Undergraduate	Bangladesh	5.2	Instagram	
1	2	22	Male	Graduate	India	2.1	Twitter	
2	3	20	Female	Undergraduate	USA	6.0	TikTok	
3	4	18	Male	High School	UK	3.0	YouTube	
4	5	21	Male	Graduate	Canada	4.5	Facebook	

Q15. How many students are there based on their 'Academic_level'?

```
1
2
3
```

```
Academic_Level
Undergraduate    353
Graduate         325
High School       27
Name: count, dtype: int64
```

Q16. Do women have a higher Addicted score than men? Calculate the average addiction score by gender.

```
1
2
3
```

```
Gender
Female    6.515581
Male      6.357955
Name: Addicted_Score, dtype: float64
```

Q17. Calculate the mean value of Avg_Daily_Usage_Hours and max value of Addicted_Score for each used platform (Most_Used_Platform).

```
1
```

Most_Used_Platform	Avg_Daily_Usage_Hours	Addicted_Score
Facebook	4.507317	5.666667
Instagram	4.872289	6.554217
KakaoTalk	4.725000	6.000000
LINE	3.250000	3.000000
LinkedIn	2.519048	3.809524
Snapchat	5.092308	7.461538
TikTok	5.346104	7.428571
Twitter	4.870000	5.500000
VKontakte	4.250000	5.000000
WeChat	4.960000	6.066667
WhatsApp	6.475926	7.462963
YouTube	4.080000	6.100000

Q18. Which country has the highest average `Addicted_Score`?

```
1
```

```
Country
Armenia           9.000000
Ecuador           9.000000
Czech Republic    9.000000
Liechtenstein     9.000000
Lebanon           9.000000
                    ...
San Marino        4.000000
Germany           3.857143
Japan             3.476190
Sweden            3.000000
Slovakia          3.000000
Name: Addicted_Score, Length: 110, dtype: float64
```

Q19. Create a new column called 'Age_Group' in which we will categorize the students based on their age. If they are <=19, the value of 'Age_Group' is Teen. If the age is bigger than 19 but less or equal than 22, 'Age_Group' will be 'Young Adult' and otherwise, 'Older student'.

```
1
2
3
4
```

Q20. Based on 'Age_Group' column categories, calculate the "Addicted_Score" mean values for each category.

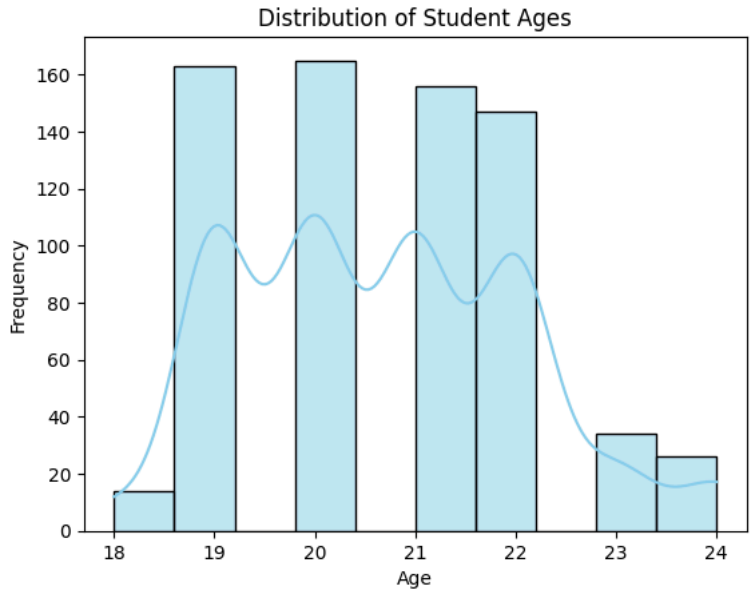
```
1
```

```
Age_Group
Older Student    5.866667
Teen             6.740113
Young Adult      6.395299
Name: Addicted_Score, dtype: float64
```

3. Visualization (4 points - 0,4 points for each question)

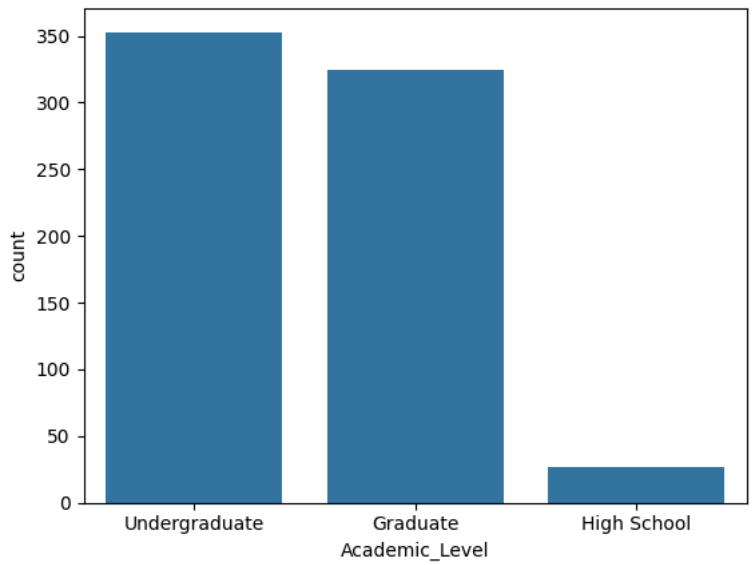
Q21 Create an histogram to show the age of students using seaborn library.

```
1
2
3
```



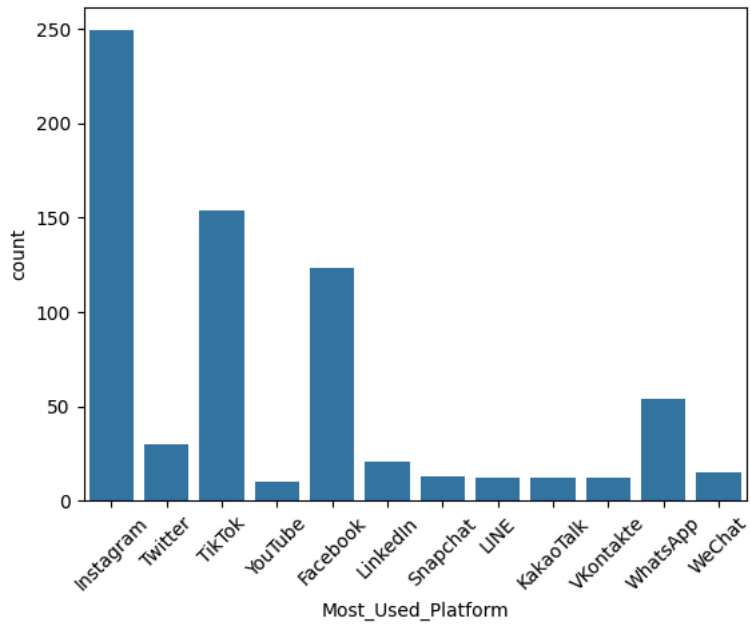
Q22 Create a graph to show the number of students based on their Academic level.

```
1
```



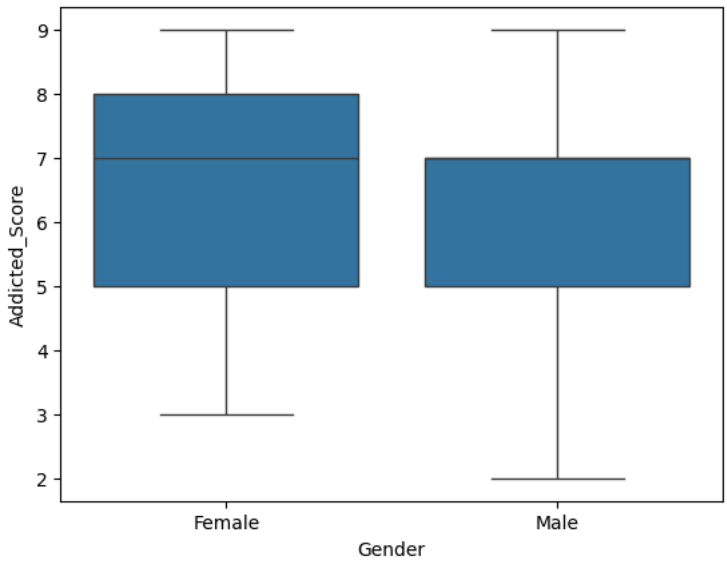
Q23 Show the number of students per platform.

```
1
2
```



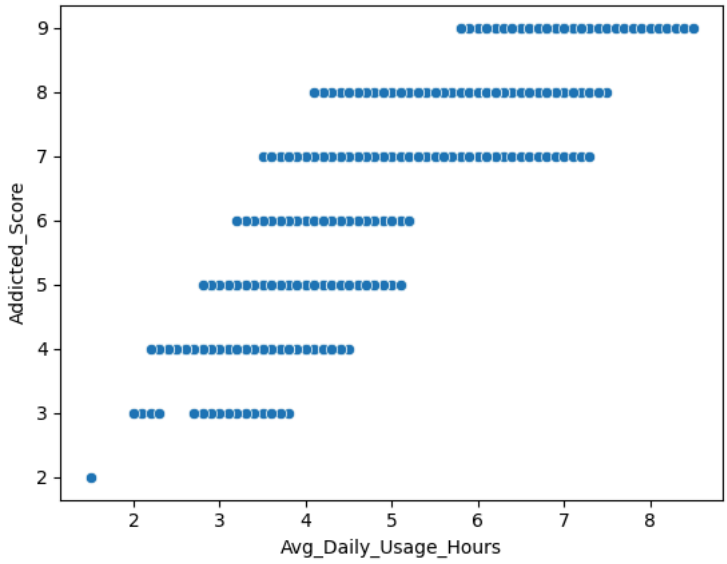
Q24 Create a boxplot to show the addiction score by gender.

```
1
2
```



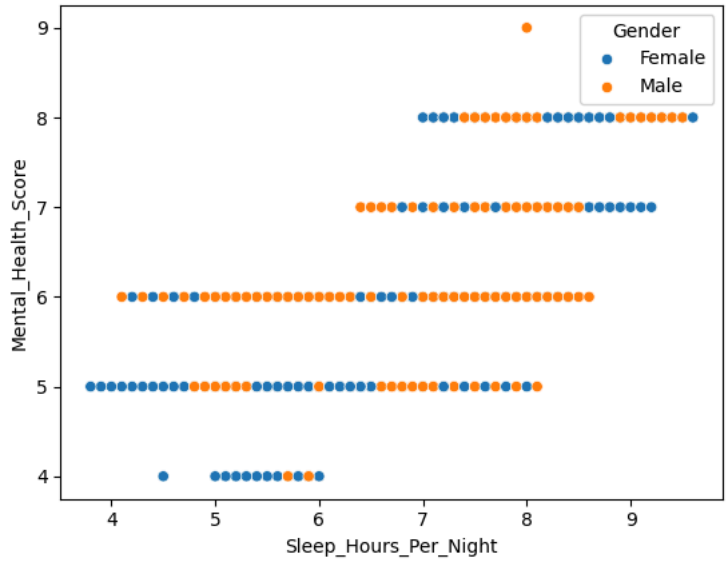
Q25 Are Avg_Daily_Usage_Hours and Addicted_Score variables related? Show it graphically.

```
1
2
```



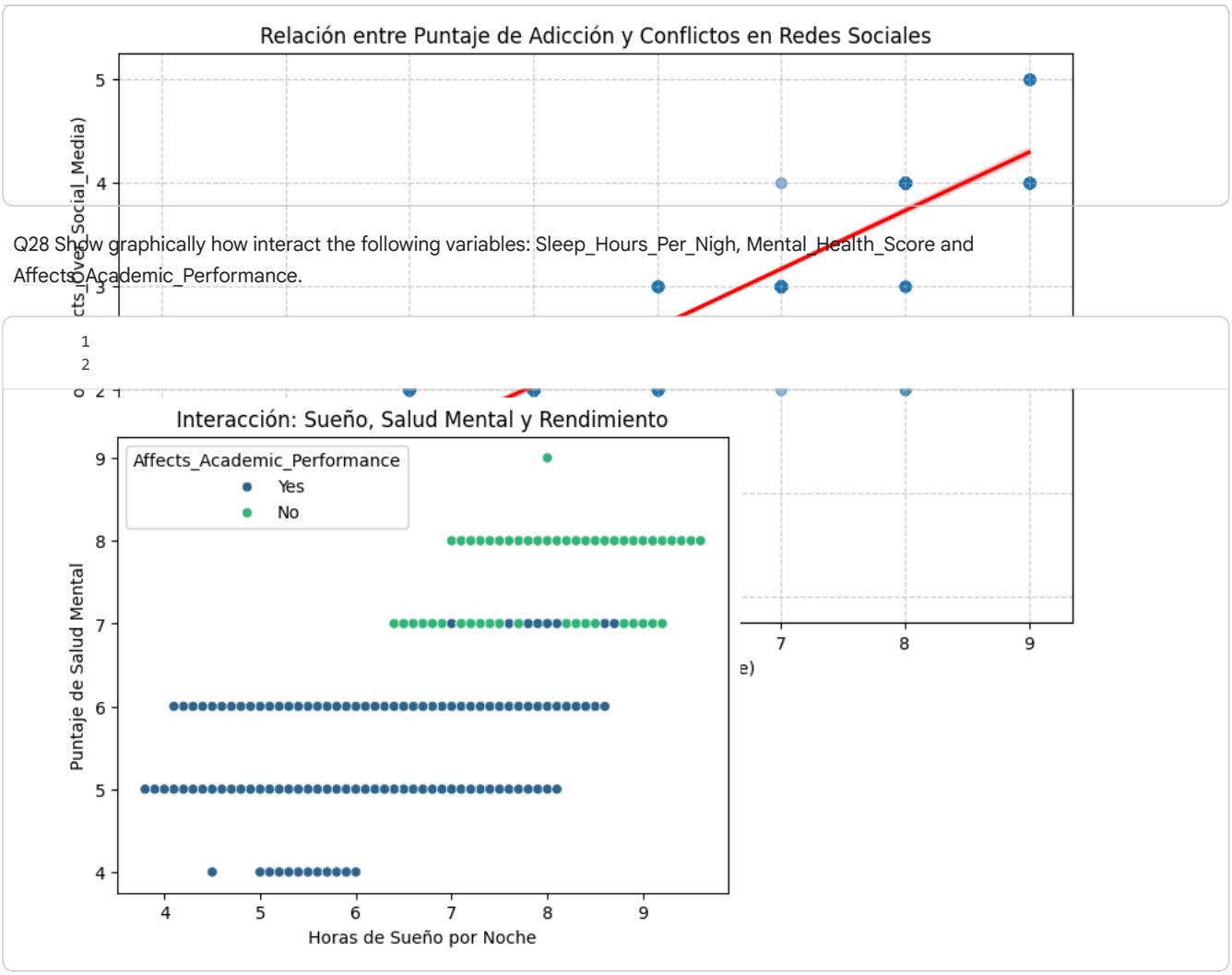
Q26 Show the relationship between Sleep Hours Per Night and Mental Health Score, using Gender as a color indicator.

```
```



Q27 Do you think the Addicted_Score affects the number of conflicts?

```
```



Q28 Show graphically how interact the following variables: Sleep_Hours_Per_Nigh, Mental_Health_Score and Affects_Academic_Performance.

Q29 Show the correlation matrix/Correlation heatmap selecting only the numeric variables.